

Taar.

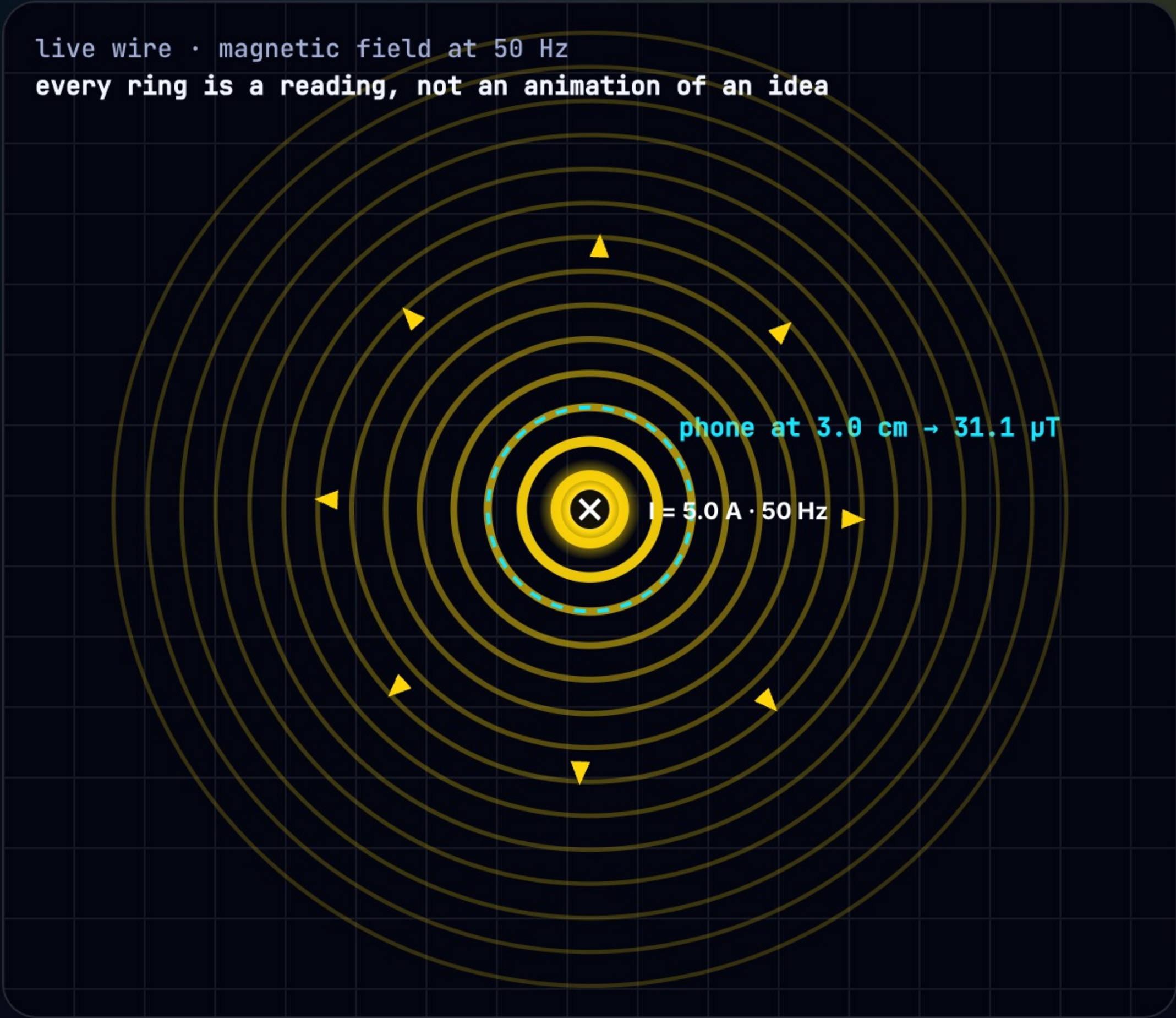
Your phone can **see electricity**.

Hold it near a wire for three seconds. It tells you whether current is flowing, how much, and whether anything is sparking. It never touches the wire, and it never touches the internet.

MAGNETOMETER → CURRENT

MICROPHONE → SPARKING

3 AI MODELS ON THE PHONE



Electrical faults start fires. The faults that start them are invisible.



A loose connection, quietly arcing

Behind a switchboard cover, a terminal that was tightened ten years ago sparks a hundred times a second. Nobody hears it. It heats, chars and one day ignites.



A circuit run over its rating

An AC and a geyser were added to a wire sized for lights. The breaker holds, the insulation slowly cooks. The bill goes up and nobody knows why.



A breaker labelled “off” that is live

Before touching a wire, an electrician needs to know it is dead. The usual method is a label on a board that may be a decade out of date, or a probe.

The instruments that would catch this — a clamp meter, a thermal camera, an arc-fault analyser — cost more than a small electrician earns in a month.

So the tools stay with large contractors, and the homes and small shops that most need checking never get checked. A family has no way at all to ask “is something wrong inside my wall?”

Two people hold the same phone.

THE ELECTRICIAN

Triage before opening a live panel

- ▶ **Confirm the switched-off circuit** is actually carrying no current before touching it.
- ▶ **Find which breaker feeds which wire** with Geiger mode, no label needed.
- ▶ **Spot a sparking joint** along a cable run and narrow down where to open.
- ▶ **Record every circuit's normal** on a board once, and compare on every later visit.
- ▶ **Hear the result spoken**, because the phone is pressed to a wall facing away.

ANYONE, AT HOME

Test your own house tonight

- ▶ **Is this wire live?** Hold the phone on it. No setup: Geiger mode clicks faster near current.
- ▶ **What is this appliance costing me?** Calibrate once with a kettle, then see amps and rupees a month.
- ▶ **Is something sparking behind the switchboard?** Measure the cable and listen for the 100 Hz pattern.
- ▶ **Which room is drawing power right now?** House View lights each room by the current in its circuit.
- ▶ **Ask in plain English:** "why did it say that?" answered by an AI that runs on the phone.

Every live wire broadcasts its current as a magnetic field.

Every sparking joint broadcasts its fault as sound.

A phone already carries both instruments.

The compass sensor is a current sensor

The magnetometer that points Google Maps north also feels the 50 Hz wobble of a mains wire. The principle a ₹5,000 clamp meter uses, without the clamp.

The microphone is an arc detector

A series arc re-ignites twice per mains cycle. Its hiss pulses at exactly 100 Hz. Nothing else in a room is locked to the grid like that.

NO PROBE

NO CONTACT

NO NETWORK

NO EXTRA HARDWARE

3 SECONDS

Only one thing in the room wobbles at exactly 50 Hz.

A wire carrying current makes a magnetic field, $B = \mu_0 I / 2\pi r$. A 5 A kettle at 3 cm gives about 33 μT , comparable to Earth's own 45 μT and far above the 0.4 μT sensor floor we measured. Strength was never the problem.

Telling it apart is. Earth's field is static, a hand drifts several μT and trembles at about 10 Hz, chargers and metal all move the needle. So Taar fits a 50 Hz sine to each axis and compares its power with the neighbouring bands, 30–48 and 52–70 Hz. That ratio is the **contrast**: room noise reads 1–3 $\times$, a kettle cord 28–62 $\times$.

Kettle OFF

Kettle ON · 5.2 A

Under 8 $\times$	no current
8 $\times$ to 13.5 $\times$	unclear, measure again
Over 13.5 $\times$	current flowing

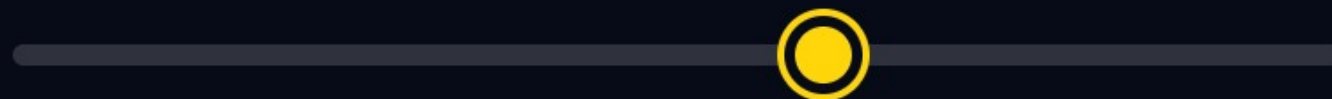


Android's default sensor rate is 100 Hz. It cannot see 50 Hz.

Everyone assumes the danger is a slow sensor. It is not. What matters is how many **distinct phases** of the 50 Hz wave the samples land on:

```
phases = fs / gcd(fs, 50)    // need  $\geq 3$  to fit sin + cos
```

Sample rate



100 Hz

DISTINCT PHASES

2

VERDICT

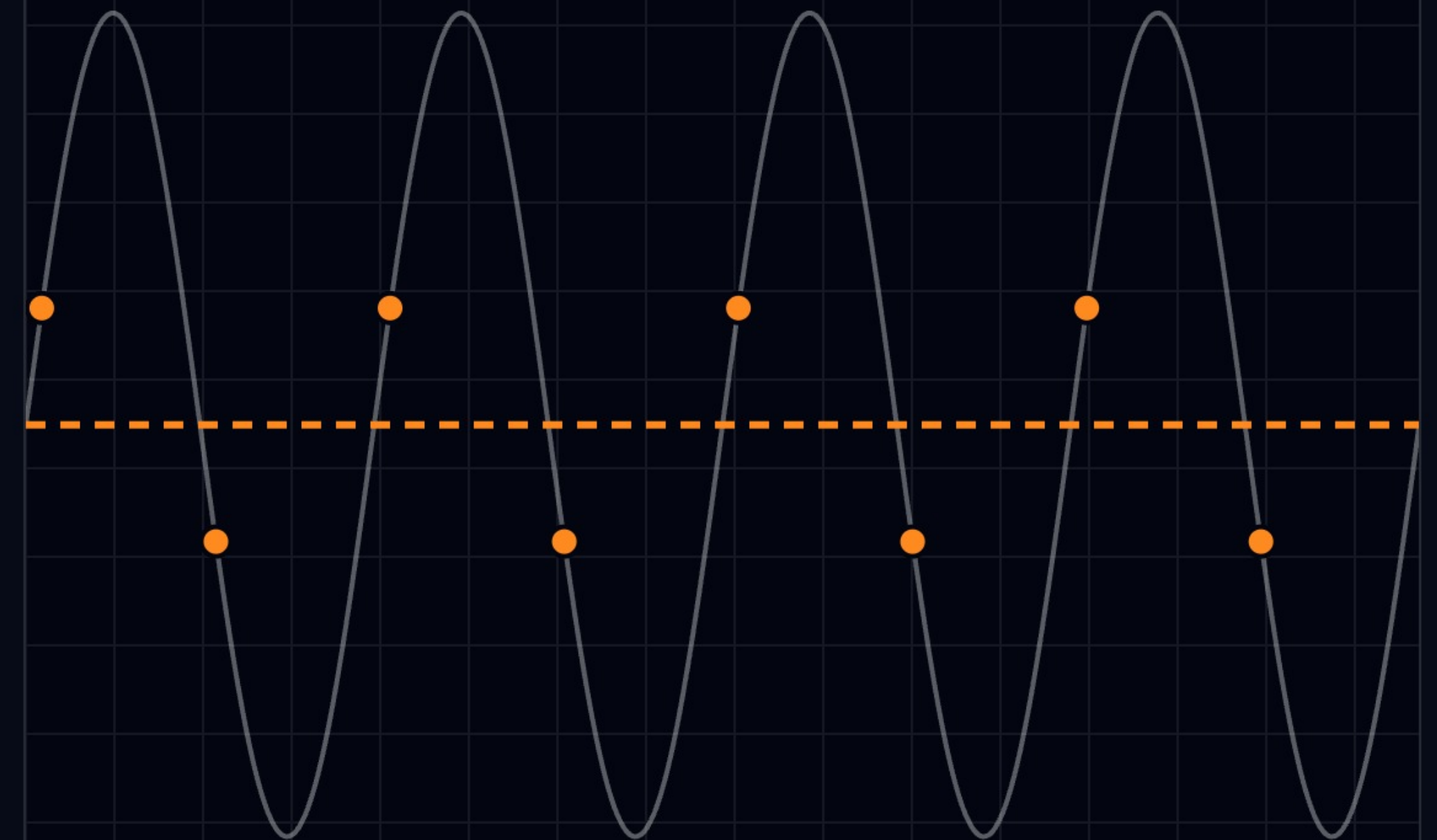
marginal

AMPLITUDE
ERROR

22%

At 50 Hz every sample lands on the same phase and the wave reads as a constant. 24 Hz recovers it to 0.3%. Locked rates below 150 Hz: 10, 20, 25, 50, 100. The fix is one call at sensor registration: `safeRate()` moves 100 $\rightarrow$ 99. The iQOO delivered 105.3 Hz over 21 phases.

50 Hz wave · dots are the samples · 100 Hz visits 2 phases of the wave



4 cycles of 50 Hz · 8 samples at 100 Hz
fitted amplitude: 0% of true (dashed)

A sparking joint re-ignites twice every mains cycle. Its hiss pulses at 100 Hz.

The microphone records 3 s. Taar keeps only the 4–16 kHz band where arc noise lives, takes the envelope, and asks how much of that envelope pulses at 100 Hz compared with 20–400 Hz. Loudness cancels out, so one threshold serves a quiet cupboard and a plant room.

Quiet room

Speech

Ballast hum

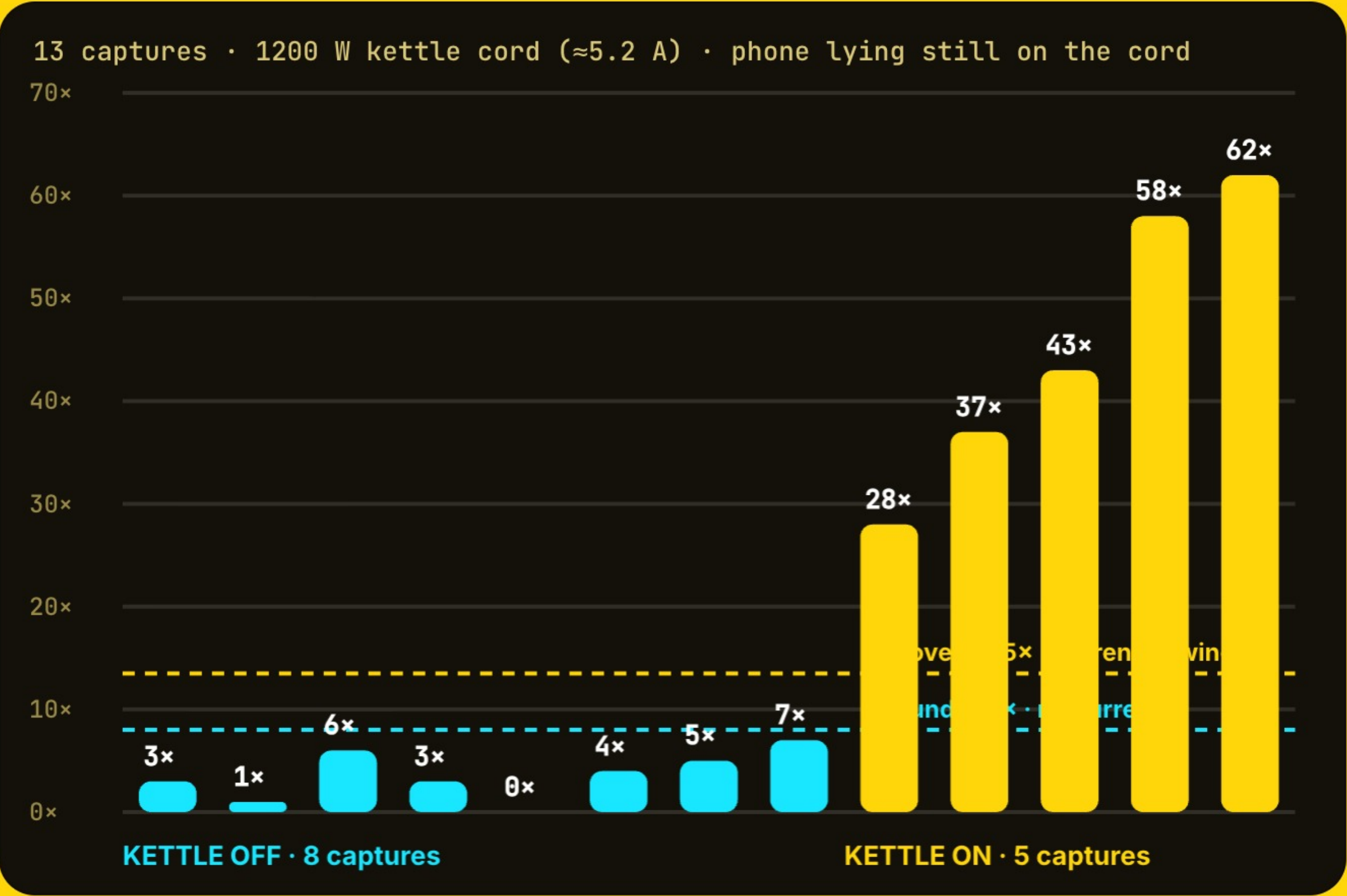
Arc

FALSE ALARMS ON	LOUDNESS DETECTOR	100 HZ DETECTOR
Ballast hum	100%	4%
Speech	100%	6%
Keys, sleeve rustle	100%	10%
Motor	100%	5%

Same lesson as the magnetometer, reached independently: do not measure energy, measure energy locked to the grid.



Same phone. Same cable. One switch.



No current, 8 captures	3× 1× 6× 3× 0× 4× 5× 7×
Kettle boiling, 5 captures	28× 37× 43× 58× 62×
Separation	factor of 4, no overlap
Sensor rate on the iQOO	105.3 Hz · 21 phases
Noise floor, two sessions	0.39–0.40 µT RMS
Earlier fridge test	9×, 16× running · 2–3× idle

Every threshold in the app comes from these numbers. Under 8× is no current, over 13.5× is current flowing, between is reported as unclear rather than guessed.

Set up once. Then three seconds per wire.

STEP 1 · ONCE

Phone check

3 s flat on a table.
Measures the real sensor rate, picks a safe one, reads the noise floor, checks the mic.

STEP 2 · ONCE PER BOARD

Name the circuits

One board, its breakers.
Type the rating printed on each, e.g. B16, so Taar can warn near the limit.

STEP 3 · ONCE PER WIRE

Record its normal

3 quick captures with the wire in its everyday state. Every later reading is compared with this.

EVERY TIME

Measure

Say if the supply is on or off. Press the phone flat on the cable, hold still for 3 s. One combined result, spoken aloud.

Why a phone check first

Because 100 Hz, the Android default, cannot see 50 Hz. Taar measures what the sensor really delivers, moves it to a safe rate if needed, and shows the number to the user. It refuses to measure on a sensor it has not checked.

Phone check

OFFLINE ARC AI ON

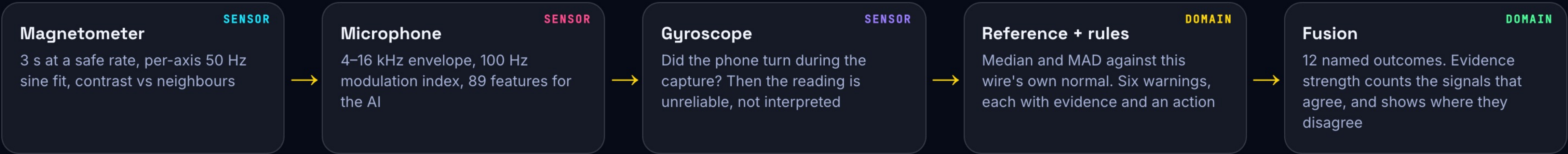
Magnetometer
105.3 Hz
21 distinct phases · safe

Background noise
0.40 μ T
phone flat on a table

Microphone
allowed
44.1 kHz

Done · Choose a circuit

Explainable sensor fusion. It says what was observed, never “safe”.



The six warnings, each ending in an action

- ! **Current on a circuit you switched off.** Stop. Mislabelled breaker or back-feed. Do not work on it.
- ! **Possible arcing / loose connection.** Isolate and have the terminations checked.
- ! **Load at or above breaker rating.** Move load or have the circuit uprated.
- ! **Load higher than usual.** If you just switched something on, expected. If not, check what was added.
- ! **Can't confirm the switched-off circuit.** Keep still and measure again. Do not assume.
- ! **Reading not usable.** Re-run the phone check and capture again.

Three rules that never bend

- ✓ **Refuse rather than guess.** No reference, an unclear signal or a moved phone gives a refusal, not a plausible number.
- ✓ **A warning only fires when every condition holds.** A partial match is a different situation, not a weak diagnosis.
- ✓ **Amperes only when calibrated.** Otherwise a relative index. A fabricated ampere figure is worse than none.

Tap “Why” on any result and every signal is listed with its verdict: supports, against, neutral or unavailable. A technician can disagree with the tool on site, because the tool showed its working.



Measure
3 s on the cable → one fused result



Reference
Each wire's own normal, 3 captures



Amps calibration
One kettle turns the index into amperes



Cost in rupees
What this wire costs a month, TSSPDCL rate



Geiger Mode
Clicks faster near a live wire. No setup.



House View
Circuits as rooms, lit by their current



Board Map
Photo of the board, a coloured dot per breaker



Cable Scan
Where along a cable the sparking is strongest



Spoken results
Hears the answer, phone facing the wall



Voice control
"Measure kitchen", "next", "why?" offline



Ask Taar AI
On-device Qwen explains any result



See What Taar Sees
Live view, signal to result, every 3 s



Teach Taar
Label a reading; thresholds learn per wire



Motion check
Gyro marks a moved capture unreliable



History
Every reading kept on the phone



Fully offline
No INTERNET permission in the manifest

Measure

Pick the circuit, say whether the supply is on or off, press the phone flat on the cable and hold still for 3 seconds. The magnetometer, microphone and gyroscope are captured together on one clock. One fused result comes back with its evidence, and a warning only when every condition for it holds.

House View

Every circuit is a room. A room glows with the current last measured on it, and the wire from the board runs at that current's pace. Drag to turn, tilt and zoom. An unmeasured room stays dark: **nothing moves unless a reading says so.**

drag to turn · rooms lit by measured current



Geiger Mode

The same 50 Hz detector over a 1.2 s sliding window, answering eight times a second. Clicks rise from 1.5 to 25 per second as the contrast climbs from 5× to 30×, on a log scale so a cable feels the same at 2 cm and 6 cm. No reference, no circuit, no setup. **Move the phone along the wall.**

move your mouse across · the wire is hidden in the wall · **No current nearby** ·
move slowly along the wall



0.0 clicks / s

The AI explains. The physics decides.

ARC MODEL · LITERT (TFLITE)

16.6 KB neural network

An MLP over 89 sound features, about 3,400 weights. Trained on 1,157 captures recorded on the iQOO itself: simulated arcs through a speaker, real not-arcs from real rooms. Held out by session, never by clip.

False alarms, rule alone	10.2%
False alarms, model	0.6%
Detection	94.8%
Self-check at every launch	golden rows · 1e-4

ASSISTANT · MEDIAPIPE LLM

Qwen2.5 · 0.5B · int8

A 550 MB language model running on the phone's CPU. Under every result it explains that reading from Taar's own evidence. In the Ask tab it answers questions about the app from 22 built-in notes, chosen by keyword overlap and shown under the answer.

May invent numbers	forbidden by prompt
May call a wire safe	flagged by regex
May decide a result	never
Temperature	0.3

VOICE · VOSK EN-IN

Offline speech, rules first

Indian-English speech recognition on the phone. A rule parser understands "measure kitchen", "next", "how much does the AC cost", "turn on geiger". Anything it does not recognise goes to Qwen to rephrase into one of those commands, so **the model can only ever pick an action from the list, never invent one.**

Conversation mode	listens again after each reply
Never by voice	mark supply off · delete

Plus a fourth learner with no training run at all: a nearest-centroid classifier fed by the technician's own labels. Mark a reading Normal, Warning or Fault on Tuesday and it is recognised on Wednesday. It rejects the unfamiliar rather than picking the nearest wrong answer, and it sits behind the rules, never in front of them.

Native Android. No server, no account, no INTERNET permission.

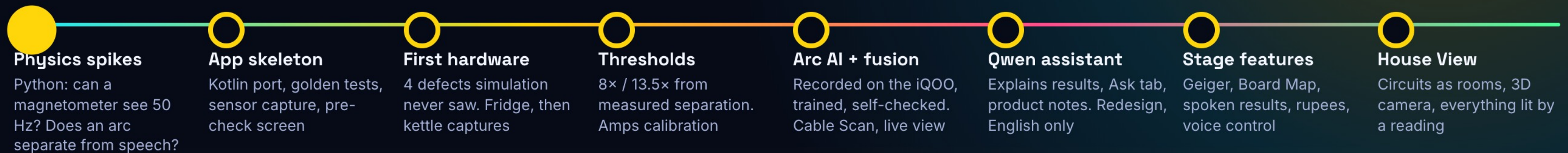
sensor/	MagCapture, MagStream, AudioCapture, MotionCapture, CaptureCoordinator on one shared clock. Per-sample timestamps kept.
dsp/	LineFrequency guard, SineFit, LombScargle, Biquad, Fft, ArcDetector, ArcFeatures, Spectrogram. Pure Kotlin, no Android.
domain/	Stats, Thresholds, Metrics, FaultCatalogue, RulesEngine, Fusion, Classifier, AmpCalibration, EnergyCost, Geiger, BoardMap, HouseView, CableScan, VoiceCommand, SpokenResult, ProductKnowledge. Pure Kotlin.
ml/	ArcModel (LiteRT interpreter + self-check), TaarAssistant (MediaPipe LLM Inference)
ui/	Jetpack Compose, Material 3. Four tabs: Home, Tools, Ask AI, History. Every task screen full-screen.
data/	Append-only, atomic, versioned line store. A capture is the one thing that cannot be retaken.

PIECE	WHAT WE USED
Language · UI	Kotlin · Jetpack Compose · Material 3 · coroutines
Android	minSdk 26 · targetSdk 35 · Gradle 9.7.1 · JDK 27
Arc model runtime	Google LiteRT 1.4.2 (TensorFlow Lite)
Language model	MediaPipe tasks-genai 0.10.35 · Qwen2.5-0.5B-Instruct q8
Speech	Vosk 0.3.47 · en-IN model · Android TTS
Research spikes	Python · NumPy · SciPy · TensorFlow (training)
Testing	golden vectors from the Python spikes · kotlinc verify.sh, no SDK needed
APK	≈24 MB · arm64 · no INTERNET permission

```
←!— AndroidManifest.xml · verifiable in thirty seconds →
<uses-permission android:name="android.permission.RECORD_AUDIO" />
<uses-permission android:name="android.permission.VIBRATE" />
←!— android.permission.INTERNET : absent. Nothing to upload. →
```

Boards live in basements with no signal, a building's circuit map is a security document, and a measurement cannot have a network round trip inside it. Every capture, model and answer runs on the phone.

Spike → measure → threshold → build → polish.



Physics spikes, before any app code

Two Python spikes answered the questions that would otherwise eat event hours. The DSP spike found the risk had been misdiagnosed: not a slow sensor, but a sample rate locked to 50 Hz. The audio spike showed a loudness detector fires on every confounder and a 100 Hz modulation index holds its 5% design point. Both produced golden CSVs the Kotlin is tested against.

What hardware taught us that simulation could not · four defects, each now a regression test

1

Two Android clocks

Sensor timestamps include deep sleep; the system clock does not. Every capture returned zero samples.

2

Magnitude erased the signal

Reducing x, y, z to one length hid a 1 μ T wobble behind Earth's 45 μ T. Each axis is now fitted separately.

3

Normalised by total variance

Hand drift buried the 50 Hz term and reported a real signal as zero. Now measured against neighbouring frequencies.

4

Guessed thresholds

The arc floor was five times too small and called a fridge an arcing fault. Every threshold is now set from captures.

What Taar does not do.

- ✕ **It does not certify.** A triage aid. It does not replace a licensed electrician, a calibrated clamp meter, an insulation tester or a statutory inspection. The result screen says so.
- ✕ **It senses current, not voltage.** A switched-off appliance on a live wire shows no current. Always use a voltage tester before touching anything.
- ✕ **It does not report amperes** unless the circuit has been calibrated against a known load. Otherwise a relative index.
- ✕ **The arc model has never heard a real arc.** It was trained on simulated arcs played through a speaker and recorded on a real phone, with real not-arcs. The pipeline retrain on a dimmer-rig recording unchanged.
- ✕ **It does not know where a cable runs inside a wall.** House View is a diagram the way a metro map is a diagram.
- ✕ **The assistant can be wrong.** It explains from the evidence, never decides, and is flagged when it sounds certain about safety.

Judges see a day of projects that claim more than they do. A team that draws its own boundary is the one they remember.

What you are about to see

0:00

Geiger sweep

Phone along the desk. Clicks rise as it crosses the kettle cord, fall as it leaves. No setup, no reference.

0:20

Kettle off → Measure

Contrast under 8×. "No current seen. Normal for this wire."
Spoken aloud.

0:40

Kettle on → Measure

Contrast jumps past 13.5×.
Amps and rupees a month appear on the calibrated circuit.
House View's kitchen lights up.

1:10

Ask it why

By voice: "why did you say that?" Qwen explains from the evidence on screen, on the phone, offline.

Safety optics, on purpose. A power strip and a domestic kettle, never an opened board on a demo floor. Deliberately arcing mains wiring is a fire hazard; the arc signal is shown from a labelled recording, and said so.

Proof on screen. Home shows the three pills: Offline, Arc AI on, Assistant on. The phone check shows the measured sensor rate the estimator was chosen from. The manifest is one tap away.

A ₹5,000 instrument, already in 600 million pockets.

- **Real arcs.** A triac-dimmer rig to record genuine line-locked arcs and retrain the same 16.6 KB model.
- **Estates.** Many boards, months of trend history, a reading that drifts on a wire before it fails.
- **More fault classes.** Phase imbalance from three conductors read together. Neutral faults. Export a report.
- **Every home.** The Geiger and House View path needs no training at all. That is the version a family installs.

